

# I Hate Mathematics

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Hi, my name is Vectra, I am a normal university student at the University of Tears, Anxiety, Deadlines, Exams and Overthinking (UTADEO) who really really hates math. My degree is systems engineering but all math doesn't matter there, I don't think that this bunch of math is ever going to be useful in life.

Me and my friends love to *parchar* in "La Brújula". The other day, a random guy came to us shouting a bunch of mathematical formulas that I didn't understand. In his madness he said that all of those formulas lead to the secret of the universe.

Although I don't like math, I do feel curious about the universe... can you help me calculate the numeric results of the formulas that he said?

You are given several queries. Each query corresponds to a specific mathematical formula. Your task is to evaluate each query and print its result.

- **ADD**( $a, b$ ) =  $a + b$
- **MUL**( $a, b$ ) =  $a \times b$
- **POW**( $a, b$ ) =  $a^b$  (using a loop, NOT built-in power)
- **FACT**( $n$ ) =  $n!$  (using a recursive function)
- **POLY**( $a, b$ ) =  $a^2 + 2ab + b^2$
- **DIST**( $x_1, y_1, x_2, y_2$ ) =  $(x_1 - x_2)^2 + (y_1 - y_2)^2$
- **AVG**( $a, b, c$ ) =  $\frac{a+b+c}{3}$
- **SECRET OF UNIVERSE**( $n, a, b$ ) =

$$\sum_{i=1}^n \left( \sum_{j=1}^n \left( \prod_{k=1}^n \left( \pi \times e^{i+j+k} + (a^2 + 2ab + b^2) + \frac{a+b+n}{3} \right) \right) \right)$$

## Program input

The first line contains an integer  $q$  ( $1 \leq q \leq 100$ ), the number of queries.

Each of the next  $q$  lines contains one of the following queries:

- **ADD** a b
- **MUL** a b
- **POW** a b
- **FACT** n
- **POLY** a b
- **DIST** x1 y1 x2 y2
- **AVG** a b c

- UNIVERSE  $n$   $a$   $b$

All values are integers in the range:

- General values:  $[-10^3, 10^3]$
- Factorial:  $0 \leq n \leq 10$
- SECRET OF UNIVERSE:  $1 \leq n, a, b \leq 30$

## Program output

For each query, print a single line with the result. Any decimal value must be rounded to two decimal places.

## Sample test cases

| Prueba #1      |           |
|----------------|-----------|
| Input          | Output    |
| 8              | 2         |
| ADD 1 1        | 4         |
| MUL 2 2        | 27        |
| POW 3 3        | 25        |
| POLY 2 3       | 25        |
| DIST 0 0 3 4   | 5         |
| AVG 3 4 8      | 120       |
| FACT 5         | 796762.03 |
| UNIVERSE 2 2 1 |           |

## Notes

- You must build complex functions using simpler ones.
- Avoid using built-in shortcuts such as `**`, `math.pow()`, or `math.factorial()`.
- You can use constant such as `math.e` for Euler number ( $e$ ) and `math.pi` for  $\pi$ .

## Important aspects

1. The exam will be completed in **2 stages**:
  - Algorithm development on paper (**3.0 points**).
    - You must write and justify the time and memory complexity of your algorithm.
    - You must identify the complexity of each function and the overall solution.
  - Implementation in the code editor (**2.0 points**).
2. The data on each query line is provided as a string. You may use:

```
1 data = input().split()
```

3. This exercise evaluates the topics covered up to **functions and loops**.
4. You must avoid lambda functions.
5. It is not necessary to use advanced data structures or external libraries.
6. Output format must match exactly.